

Stablecoin StressBench: An Execution-Aware Benchmark for Stablecoin Dislocations

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Abstract

Quoted stablecoin price gaps are not trades: we call dislocations where a quoted basis exceeds a threshold *optical*, reserving *executable* for the subset where net profit survives VWAP slippage, taker fees, and settlement latency after routing through a live order book. Building **Stablecoin StressBench** on the March 2023 USDC/SVB event, we find that 34.3% of minutes show a > 10 bps optical dislocation but only 2.88% are executable: a **12 $\times$ price-to-execution gap** (block bootstrap 95% CI: $8\times$ – $24\times$), with no calm-trained model closing it and the hindsight oracle exceeding +225 bps net on the executable arbitrage task. Cross-mechanism meta-labeling trained on Terra/LUNA achieves +82.5 bps on the SVB test split (50.8% of oracle returns), demonstrating that order-book execution microstructure transfers across structurally distinct stress mechanisms. A three-way RL diagnostic isolates why: unconditioned PPO-GRU degenerates to NoTrade at 2.3% positive density; conditioned PPO-GRU at 17% density trades 919 times but earns –29.2 bps; only meta-labeling’s explicit positive-label supervision reaches +82.5 bps—establishing that density is necessary but not sufficient.

Keywords

stablecoin arbitrage, market microstructure, execution-aware labels, financial benchmarks, cryptocurrency

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1 Introduction

During stablecoin stress events, peg fragmentation creates cross-venue price gaps that appear to be arbitrage opportunities. In practice, VWAP slippage, taker fees, and CEX-to-CEX settlement latency jointly eliminate most apparent profit. We call dislocations where the quoted basis exceeds a threshold *optical*, and those where net profit survives execution costs *executable*.

Prior machine-learning work on stablecoin predictability uses price-based labels to define “profitable” windows, which conflates the optical and executable layers and systematically overstates how predictable actual trading opportunities are. **Stablecoin StressBench** is designed to close this gap. It provides per-minute execution-aware labels that separate three dislocation layers: (1) an *optical* dislocation (quoted basis exceeds a threshold), (2) an *executable* dislocation (net profit is positive after VWAP execution, fees, and settlement at a specified notional), and (3) a *predictable* dislocation (a model correctly identifies the window ex ante). The distance between these layers is the paper’s core measurement.

We build the benchmark on the March 2023 USDC/SVB stress event, where Silicon Valley Bank’s collapse triggered a brief USDC depeg. Using Binance L2 order-book data¹ and a full VWAP book-walking model, we find that 34.3% of 1-minute windows show a > 10 bps price basis but only 2.88% remain net-profitable at \$10K notional: a 12 $\times$ price-to-execution gap. The headline result is robust to buy-leg proxy error: a 20% depth haircut moves the executable rate from 2.88% to 2.40% and widens the ratio from 12 $\times$ to 14 $\times$, with the oracle remaining above +130 bps. We then construct a model ladder spanning rule-based thresholds, statistical baselines, gradient-boosted classifiers, sequence models (GRU), and meta-labeling filters, evaluating each model’s ability to identify the 2.88% of executable windows ex ante. Every model trained on calm-control data loses money on executable arbitrage tasks; the hindsight oracle gap exceeds 260 bps. Cross-mechanism meta-labeling, trained on a different stress event (Terra/LUNA 2022), is the only approach that achieves positive net returns (+82.5 bps, 50.8% oracle capture).

Contributions.

- (1) **Execution-aware benchmark and labels:** per-minute labels with documented order-book depth provenance, VWAP book-walking, fees, and a hindsight oracle ceiling; accompanied by an 18-event stress taxonomy (seven mechanism classes, 2020–2023) that functions as data-tier governance specifying permitted claims per event, and an oracle gap decomposition into FP cost, timing, and sizing components. (§4, §3.3; Tables 4, 3, 5)
- (2) **Cross-mechanism transfer:** meta-labeling trained on Terra/LUNA achieves +82.5 bps (50.8% oracle capture) on the SVB test split—to our knowledge the first demonstration that order-book execution microstructure transfers across structurally distinct stress mechanisms, on the single cross-mechanism pair available with route-complete test data. (§6.5, Table 8)
- (3) **RL diagnostic—density necessary, supervision sufficient:** conditioned PPO-GRU at 17% positive density earns –29.2 bps vs. meta-labeling’s +82.5 bps at identical density, isolating explicit positive-label supervision as the additional requirement beyond RL reward optimisation. (§6.8, Table 10)

2 Related Work

Stablecoin stress. Uhlig [1] analyses the Terra/LUNA collapse, which we use as a cross-mechanism validation split: the algorithmic redemption spiral provides a structurally different stress trigger than the SVB bank shock, making it a clean out-of-distribution test for cross-mechanism transfer in execution microstructure. Griffin and Shams [2] find that quoted prices can reflect non-fundamental demand, informing our decision to separate optical from executable dislocations. Hernandez Cruz et al. [21] study USDC-USDT DEX

¹The BTC-USDC buy leg uses kline-proxy depth for the SVB window; all other legs use real L2. The 20% depth haircut sensitivity in §6.1 bounds this approximation.

liquidity during the SVB collapse, motivating our execution-layer analysis.

Limits to arbitrage and crypto fragmentation. Shleifer and Vishny [4] and Gromb and Vayanos [5] establish that arbitrage is limited by capital constraints and execution frictions; we operationalise this at the per-minute label level. Makarov and Schoar [7] show that cross-venue crypto price gaps are not executable profits once latency and depth are modelled, motivating our $12\times$ price-to-execution gap. Hautsch et al. [3] quantify settlement latency as a first-order cost; we implement this as the fixed $\delta=5$ bps penalty in Equation 1. Cont and Kukanov [6] establish VWAP execution price as the correct profitability object; our book-walking model in §4.2 implements this directly. Gogol et al. [18] introduce a Maximal Arbitrage Value (MAV) between a USDC-ETH AMM pool and Binance, the closest prior analogue to our oracle ceiling; our oracle extends this to route-level CEX simulation, stress-event dislocations, and model-evaluation benchmarking.

Financial ML and benchmarks. López de Prado [8] introduces meta-labeling as a technique for filtering a primary signal with a secondary model that learns when the primary signal produces true versus false positives. We adopt this architecture in §6.5, using a LightGBM secondary model conditioned on primary-signal fires to learn the microstructure signature of executable windows. The cross-mechanism variant, training on Terra/LUNA and evaluating on SVB, extends meta-labeling to out-of-distribution stress events, which to our knowledge has not been done before. Cintra and Holloway [9] apply BOCPD [15] to detect the Terra depeg 12 hours early from Curve AMM pool reserves. We test BOCPD as a stress-regime pre-filter and find AUROC 0.23 on CEX cross-quote data: AMM reserves accumulate gradually, but CEX basis is mean-reverting, explaining why their result does not transfer. Cintra and Holloway also demonstrate cross-event transfer (2022 UST training, March 2023 USDC depeg), but via AMM reserve depletion signals rather than order-book execution features, with no oracle evaluation. Moallemi and Wang [16] and Ning et al. [17] apply RL to optimal order execution in equity markets, establishing the finite-horizon MDP framing we adopt in §5.2. Mohl et al. [22] train IPPO on roughly 400 million real limit-order-book orders for a two-agent execution-and-market-making setup, showing GPU-scale multi-agent RL is feasible on production LOB data. Olby et al. [23] build a reactive Simudyne simulator that decomposes slippage into market impact and execution risk, evaluating RL timing policies against an Almgren–Chriss efficient frontier; their reactive environment is the closest prior analogue to the setting our RL diagnostic targets. StressBench contributes the complementary finding: in a non-reactive historical-replay setting with structurally rare positive labels, density and explicit positive-label supervision are the binding constraints—a pre-condition any reactive extension must satisfy. No prior stablecoin-stress benchmark jointly provides one-minute executable labels, route-level depth and fee/settlement modelling, a hindsight profit ceiling, and cross-mechanism transfer evaluation. Gogol et al. [18] bound AMM arbitrage without benchmark labels; Cintra and Holloway [9] transfer AMM depletion signals, not executable microstructure; Adams et al. [19] measure slippage without oracle evaluation; and Wu and Liu [20] study spillovers, not

Table 1: Route-leg depth provenance.

Leg	SVB Mar 2023	2024 windows	
Buy BTC-USDC	kline-proxy [†]	real L2 (futures)	[†] BTC-USDC
Cross USDC-USDT	real L2 (Mar 12–20)	real L2 (futures)	
Sell BTC-USDT	real L2 (futures)	real L2 (futures)	
Ref BTC-USD	candle-proxy	candle-proxy	

perpetual listed Jan 2024; all 2022 windows kline-proxy.

Table 2: Dataset splits (56,134 1-min bars).

Split	Event	Min.	Pos.
Train	Calm control ($\times 3$)	28,776	0.00%
Validation	Terra/LUNA May 2022	11,526	2.30%
Test	USDC/SVB Mar 2023	15,832	2.88%

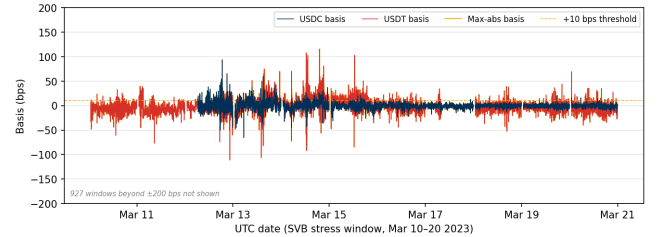


Figure 1: Cross-quote basis (bps) during the SVB test window (Mar 10–20 2023). USDC basis (navy): 12.4% of minutes > 10 bps; USDT basis (red): 27.4%; max-abs (gold): 34.3%. The price-to-execution gap is $12\times$.

execution simulation. Stablecoin StressBench is the first to combine all five.

3 Benchmark Design

3.1 Data Sources and Instruments

The final dataset contains 56,134 1-minute bars across six windows (Table 2). Price features come from Binance Vision (klines, aggTrades) and Coinbase REST candles. Net-profit labels use Binance BTC-USDC and BTC-USDT depth plus Coinbase BTC-USD as the reference leg.

Every depth record is tagged with a provenance field (Table 1). The sell leg (BTC-USDT) uses real L2 futures bookDepth for all 2023–2024 windows. The cross leg (USDC-USDT) uses real L2 from 2023-03-12 onward. The buy leg (BTC-USDC) uses kline-proxy for the SVB window (BTC-USDC perpetual did not exist on Binance until 2024) and real L2 for 2024 windows. To bound this proxy error, we apply a 20% depth haircut to the kline-proxy buy leg: the executable rate falls from 2.88% to 2.40%, the ratio widens from $12\times$ to $14\times$, and the oracle stays above $+130$ bps. The headline result is robust to buy-leg proxy error.

3.2 Event-Based Split Design

Splits are defined by event identity (Figure 1). No model trained on the train split has seen stress dynamics; threshold calibration on validation uses an independent stress event. Labels are constructed by aligning each bar’s outcome to the observation window ending at $t+h$, with explicit checks confirming no future information leaks into the input features.

Table 3: Data tiers, mechanisms, and execution frictions.

Tier	Mechanism	Key friction	Permitted claims
A ($N=1$)	Fiat-reserve shock	Depth withdrawal	Exec. gap; oracle gap; P&L
B ($N=11$)	Alg., exchange, DeFi	AMM slippage; venue seg.	Price magnitude (est.)
C ($N=5$)	RWA, niche	Thin market	Taxonomy only

3.3 Historical Stress Taxonomy: Data-Tier Governance

The 18-event taxonomy (2020–2023) is a benchmark governance artefact: it specifies which events carry sufficient data quality for execution-grade claims, and which are restricted to price magnitude estimates or mechanism classification only. It is shipped with the benchmark to foreclose over-interpretation of results, not as an empirical finding in its own right. Seven mechanism classes are covered: *algorithmic/reflexive* (FEI, IRON/TITAN, MIM, UST, USDD); *fiat-reserve bank shock* (USDC/SVB); *regulatory winddown* (BUSD); *exchange credit* (Celsius/3AC, HUSD, FTX [10]); *DeFi pool imbalance* (Curve/UST, USDT/Curve, USDC/DAI); *collateral liquidation* (DAI Black Thursday); *RWA/niche* (Acala aUSD, USDR).

Data availability governs permitted claims (Table 3). Tier-A events (USDC/SVB test) support execution-gap, oracle-gap, and model evaluation claims. Tier-B ($N=11$, OHLCV/pool data) support price magnitude estimates only. Tier-C ($N=5$, partial/DEX-only) support mechanism taxonomy only.

3.4 Benchmark Versioning

StressBench v1.0 ships with the SVB test split; near-term Tier-A expansion targets are the USDC recovery window (Mar 15–31 2023) and 2024 BTC-USDC windows where real-L2 eliminates the kline-proxy. Release format: versioned Hugging Face dataset under CC-BY 4.0.

3.5 Feature Sets

Four nested feature sets isolate the marginal information value of

	price_only	5 cross-quote basis cols
	price_plus_book	+7 microstructure (spread, depth, imbalance)
each signal layer:	price_book_frag	+3 cross-venue fragmentation cols
	price_book_settle	+7 on-chain settlement proxies

4 Execution-Aware Labels

Labels operationalise the three-layer framework introduced in §1. The price-to-execution gap measures Layer 1 divided by Layer 2. The oracle gap measures Layer 3 against Layer 2 in hindsight. This section details label construction.

4.1 Cross-Quote Basis and Labels

$$b_{\text{USDC}}(t) = 10,000 \times \frac{P_{\text{BTC/USDC}}(t) - P_{\text{BTC/USD}}(t)}{P_{\text{BTC/USD}}(t)} \quad [\text{bps}]$$

$b_{\text{max}} = \max(|b_{\text{USDC}}|, |b_{\text{USDT}}|)$. Basis label: $y_{\text{basis}}(t) = 1[|b_{\text{USDC}}(t+h)| > \tau]$. Primary task: basis_usdc_1m_gt10bps.

4.2 VWAP Execution and Net Profit Labels

$$\text{net}(q, t) = 10,000 \times \left[\frac{\text{VWAP}_{\text{sell}} - \text{VWAP}_{\text{buy}}}{P_{\text{ref}}} - f_{\text{buy}} - f_{\text{sell}} - \delta_{\text{settle}} \right] \quad [\text{bps}] \quad (1)$$

$$y_{\text{arb}}(q, h, t) = 1 \left[\max_{s \in [t, t+h]} \text{net}(q, s) > c \right] \quad (2)$$

P_{ref} is the Coinbase BTC-USD reference price; $f_{\text{buy}}, f_{\text{sell}}, \delta_{\text{settle}}$ are dimensionless fractions (e.g., 4 bps taker fee $\Rightarrow f_{\text{buy}} = 0.0004$). Parameters: $c=10$ bps, taker fee 4 bps/side, $\delta=5$ bps. Primary economic task: executable_arb_q10000_5m. The oracle trades every minute where $\text{net}(q, t) > c$ in hindsight, setting the performance ceiling no deployable model can exceed.

5 Models and Evaluation

5.1 Model Ladder

The ladder tests whether stronger models close the oracle gap, with each rung testing a distinct hypothesis.

Economic anchors. NoTrade (0 bps) and Oracle (hindsight ceiling).

Rule baselines. PriceBasis{10, 25}bps and GrossArbThreshold.

ML classifiers. Logistic, Lasso, Random Forest, XGBoost, LightGBM on four nested feature sets.

Expected-net-profit regressor. Regresses $\hat{y}=\text{net}(q, t)$ directly; failure implies a structural gap.

Meta-labeling filter. Secondary LightGBM learns $\Pr[\text{net} > c \mid |b_{\text{USDC}}| > 10 \text{ bps}]$.

Sequence models. GRU and MLP-Window on 30-min sliding windows (§6.4).

5.2 RL Execution Policy Baseline

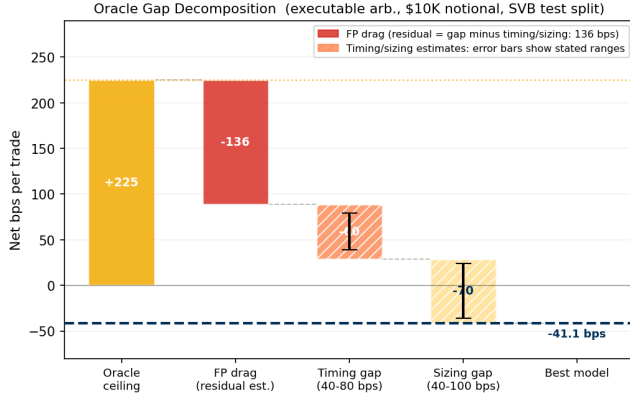
We frame the entry-timing problem as a finite-horizon MDP. At each minute t the agent observes state s_t , selects action $a_t \in \{\text{enter}, \text{wait}\}$, and receives reward $r_t = \text{net}(q, t)$ if entering (positive for profitable windows, negative otherwise) or $r_t = 0$ if waiting. The state s_t is the 30-minute look-back window of price_plus_book features, the smallest feature set that delivers meaningful microstructure lift over price-only signals (see §6.5 for SHAP evidence). This feature set is available in real time without settlement-latency dependency. We train a PPO agent with a GRU policy network (one hidden layer, 64 units, 30-step look-back) using the cross-mechanism protocol: training on Terra/LUNA stress windows only, evaluation on the USDC/SVB test split without retraining. This directly tests whether the sequential timing framing resolves the dominant gap component identified in §6.7.

5.3 Calibration and Evaluation

Threshold calibration on validation split: $\theta^* = \arg \max_{\theta} \sum_{i: \hat{p}_i > \theta} \text{net}(q, t_i)$ subject to ≥ 25 trades. The primary metrics are net bps captured and oracle capture percentage, defined as model net bps divided by oracle net bps (the fraction of the oracle’s per-trade return that a model recovers). Secondary metrics are AUROC and AUPRC. NoTrade at 0 bps is the economic floor.

Table 4: Price-to-execution gap (\$10K, 1-min horizon, test split).

Thresh.	$P_{\max}$	P_{USDC}	Exec.	Ratio
0 bps	95.86%	82.96%	3.34%	29×
5 bps	61.99%	25.32%	3.03%	20×
10 bps	34.33%	12.45%	2.88%	12×
25 bps	9.54%	6.48%	2.62%	3.6×

**Figure 2: Oracle gap decomposition (\$10K notional). Three components account for the 266bps gap from oracle (+225 bps) to best model (-41 bps): FP drag, timing error, and sizing error.**

Three claim boundaries apply: (1) the BTC-USDC buy leg uses kline-proxy depth for the SVB window; (2) oracle-gap results apply to the SVB test split ($N=15,832$) only; (3) cost parameters are $c=10$ bps, 4 bps taker fee per side, $\delta=5$ bps.

6 Results

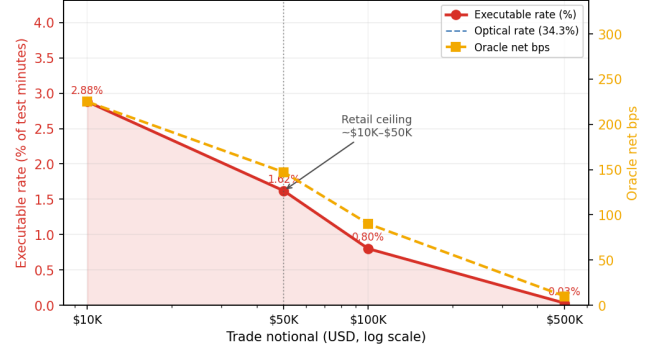
6.1 Price Gaps Are Common, Execution Is Rare

At 10 bps, 34.3% of test minutes exceed the max-basis threshold; only 2.88% yield net profit, a **12×** price-to-execution ratio (block bootstrap 95 % CI: $8\times-24\times$, $B=10,000$, 60-min blocks). The gap persists at all tested thresholds (Table 4) and is robust across cost regimes: at the worst-case setting (10 bps settlement, high fees), the executable rate falls to 2.40% and the ratio widens to 14 $\times$. At \$50K notional the executable rate falls to 1.62%; at \$500K it reaches 0.03%, confirming that book depth, not fees, prices out institutional participants. At institutional scale, book depth rather than fees is the binding constraint (0.03% executable rate at \$500K), consistent with the depth-based limits-to-arbitrage arguments in §2.

Claim boundaries and data-tier scope are stated in §5.3; the Terra/LUNA cross-mechanism check (5.9 $\times$ ratio; §6.6) provides directional evidence that the gap is not SVB-specific.

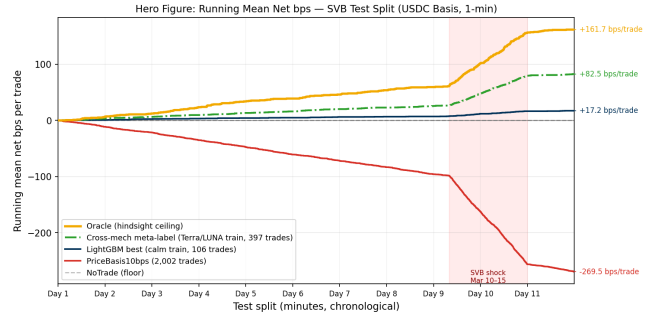
6.2 Oracle Gap: Profitable Windows Exist but Only in Hindsight

The oracle earns +162.2 bps on the basis task and +225.0 bps on the executable arbitrage task (Table 5); Figure 4 shows the full performance hierarchy.

**Figure 3: Notional scaling (SVB test split). The executable rate collapses from 2.88% at \$10K to 0.03% at \$500K: book depth, not fees, is the binding institutional constraint.****Table 5: Oracle gap summary (USDC/SVB test split[§]). * =only non-oracle tabular result above zero; ‡ =sequence model (30-min window).**

Task / Model	Oracle	Best model	Gap	AUPRC
basis_usdc_1m	+162.2 bps	+17.2 bps*	145 bps	0.253
exec_arb_q10k_5m	+225.0 bps	-41.1 bps	266 bps	0.060
exec_arb_q50k_5m	+147.1 bps	-113.0 bps	260 bps	0.063
GRU (seq) [‡] , basis_usdc_1m	+162.2 bps	-239.0 bps	401 bps	—

*lgbm@price_book_settle, 106 trades. ‡GRU 30-min window (\$6.4). §Kline-proxy buy leg; 20% haircut sensitivity in §6.1.

**Figure 4: Running mean net bps per trade (USDC/SVB test split). Oracle (gold) +162.2 bps; cross-mech meta-label (green) +82.5 bps; LightGBM (navy) +17.2 bps; PriceBasis10bps (red) -269.5 bps; NoTrade (grey dashed) 0 bps. Shaded region: SVB shock window.**

6.3 The Model Ladder: Structural Oracle Gap

Table 6 shows the full model ladder with loss attribution. Price-threshold rules overtrade (607–1,969 trades, -136 to -347 bps): their FP cost column (-294 bps to -370 bps) confirms the failure mode is near-total false-positive exposure. LightGBM achieves +17.2 bps (106 trades, AUROC 0.72), the only non-oracle result above zero, but 145 bps below the oracle. The ExpNetProfitRegressor reaches -73.8 bps despite directly targeting the economic criterion: the gap is structural, not a label-mismatch artefact.

Ranking accuracy does not imply positive returns. At 2.88% positives, precision-recall curves expose the imbalance problem directly

Table 6: Tabular model ladder (basis_usdc_1m_gt10bps, test split). Hit%: fraction of model trades that were net-profitable. [†]ceiling; *only result above zero. Sequence and RL models in Tables 7 and 10.

Model	Feat.	Net bps	Trades	AUROC	Mean bpswhen wrong	Hit%
Oracle [†]	—	+162.2	315	—	—	100%
lgbm*	all	+17.2	106	0.717	−65.2	33%
logistic	all	−75.8	648	0.143	−75.8	0%
gross_arb	px	−136.4	607	0.531	−150.9	4%
price_10	px	−270.0	1969	0.833	−294.6	6%
price_25	px	−347.3	1025	0.746	−370.7	5%
no_trade	—	0.0	0	—	—	—
ExpNetProfit	px	−73.8	537	—	−89.1	3%

px = price_only; all = price_book_settle.

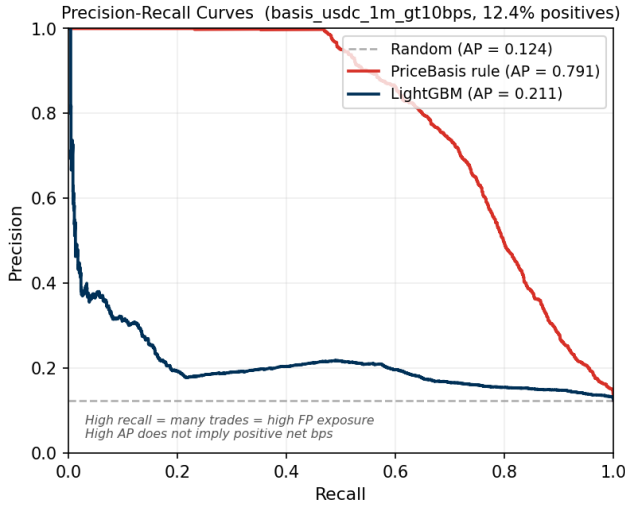


Figure 5: Precision-recall curves (basis_usdc_1m_gt10bps, SVB test split). High AP does not imply positive net bps: PriceBasis achieves AP 0.55 while losing −270 bps.

(Figure 5). PriceBasis10bps achieves AP 0.55 and AUROC 0.83 while losing −270 bps; LightGBM achieves AP 0.25 and AUROC 0.72 as the sole non-oracle model above zero. High AP or AUROC is necessary but not sufficient for positive returns. High AUROC with low Hit% reflects the threshold regime: the calibrated threshold sits in a high-recall, low-precision region where most trades are false positives, even though the model’s overall ranking is correct.

6.4 Sequence Models: Confirming the Structural Oracle Gap

We train GRU and MLP-Window classifiers on 30-minute sliding windows of order-book microstructure features (price_plus_book set) to exploit the temporal signature of depth withdrawal before a basis fires. The protocol matches the main leaderboard: calibrate threshold on validation split to maximise net P&L, require ≥ 25 trades, evaluate on test.

On basis_usdc_1m_gt10bps, evaluated on SVB test data (Table 7):

Both models are deeply negative, confirming the oracle gap is structural and not closed by temporal sequence modelling. AUROC 0.80 for GRU combined with −239 bps net illustrates the ranking-vs-returns disconnect: the calibrated threshold admits

Table 7: Sequence model results (basis_usdc_1m_gt10bps, SVB test data, test split). Oracle gap = +162.2 bps oracle minus model net bps.

Model	Net bps	Trades	AUROC	Gap
GRU (30-min)	−239.0	656	0.798	401 bps
MLP-Window (30-min)	−340.7	1,052	0.648	502 bps

Calm-control trained (0% positive examples); threshold calibrated on validation.

twice as many trades as the oracle because calm training produces zero positive stress examples. Cross-mechanism meta-labeling (§6.5) bypasses this by training on Terra/LUNA stress dynamics, achieving +82.5 bps.

6.5 Cross-Mechanism Transfer: Benchmark Generalisation Evidence

Meta-labeling result. Calm-control training yields zero meta-positives (0% positive rate), so we retrain on Terra/LUNA (1,559 primary fires, 265 meta-positives, 17.0% rate) and evaluate cross-mechanism on SVB. All fitting and threshold selection use Terra/LUNA only; no SVB labels are observed before final evaluation. With price_plus_book features the filter achieves +82.5 bps (397 trades, 50.8% oracle capture) vs. zero with calm training (Table 8). This is consistent with execution microstructure that transfers across stress mechanisms: the same order-book features that predict profitable windows during an algorithmic collapse (Terra/LUNA) also predict them during a fiat-reserve shock (SVB). The meta-model’s dominant features mirror the main SHAP ranking: ask-side depth at 10 bps and bid-ask spread are the leading microstructure signals in both calm-trained LightGBM and the Terra-trained meta-model secondary classifier. The structural explanation: CEX market makers withdraw depth under uncertainty regardless of trigger, because the cause is opaque at the order-book timescale, making depth withdrawal a consistent cross-mechanism precursor whose feature signature transfers while the magnitude differs by event (5.9× Terra/LUNA vs. 12× SVB). We are not aware of prior work demonstrating cross-mechanism transfer for execution-aware stablecoin models.

Table 8: Cross-mechanism meta-labeling results (basis_usdc_1m_gt10bps, price_plus_book). Oracle capture = meta net bps ÷ oracle net bps.

Training split	Net bps	Trades	Oracle cap.
Oracle (ceiling)	+162.2	315	100.0%
Terra/LUNA (cross-mech.)	+82.5	397	50.8%
Calm-control (baseline)	0.0	0	0.0%

signal $|b_{USDC}| > 10$ bps; threshold calibrated on validation.

Regime detectors and false-positive diagnosis.

Regime detectors. CUSUM achieves near-perfect recall (0.9995) at 83.5% false alarm rate; EWMA achieves 86.9% accuracy at 0.65% FAR with 0.95% recall; BOCPD achieves AUROC 0.229 on CEX cross-quote data. This last result contrasts with Cintra and Holloway [9], who found BOCPD detected the Terra depeg 12 hours early from Curve pool reserves. The difference is the signal: AMM reserves reflect LP withdrawal, which accumulates gradually, while CEX

Table 9: Cross-mechanism comparison.

Metric	USDC/SVB (Tier-A)	Terra/LUNA (val.)
Minutes	15,832	11,526
Mechanism	Fiat-reserve	Alg./reflexive
Price > 10 bps	34.3%	13.5% [†] Kline-proxy
Executable	2.88%	2.30%
P-to-E ratio	12×	5.9×
Oracle bps	+225.0 bps	+88 bps [†]

labels; directional only.

cross-quote basis is noisier and mean-reverting, making change-point detection unreliable on CEX prices.

False-positive structural diagnosis. PriceBasis10bps produces 2,002 trades; 581 (29%) are FPs at mean -38.7 bps. The cause is route-direction mismatch: true-positive windows have a USDC basis of 344 bps (genuine discount), while FP windows have 0.56 bps—triggered by USDT dislocations where the USDC route is not simultaneously profitable. FP windows show *higher* bid depth (72,805 vs. 59,961 BTC units), ruling out thin books; the failure mode is direction, not depth. LightGBM’s 33% hit rate (vs. 6%) shows microstructure conditioning partially diagnoses mismatch, but 67% FP remains a timing gap.

SHAP feature importance. SHAP analysis of LightGBM shows price features carrying 94.8% of total importance; ask-side depth at 10 bps and bid-ask spread contribute just 3.4% combined. The implication is causal: a calm-trained model learns calm price dynamics, not stress execution structure—the microstructure features are present in the input but contribute negligibly because the calm training split contains no profitable stress windows from which the model could learn their relevance. This is why the oracle gap is structural rather than a label-mismatch artefact: the features needed to identify executable windows are available but the training distribution renders them uninformative. Cross-mechanism training on Terra/LUNA is the fix: the meta-model sees stress microstructure at 17% positive density and learns the depth-withdrawal signature that calm training cannot, tying the model ladder, oracle gap, and transfer result into a single causal account.

6.6 Terra/LUNA Replication: Pattern Generality

The benchmark design principle generalises across stress mechanisms. The Terra/LUNA validation split ($N=11,526$, May 2022) uses kline-proxy depth (Tier-B) but replicates the core pattern: 13.5% price-basis > 10 bps, only 2.30% executable, a 5.9× ratio vs. 12× for SVB. The oracle earns +88 bps on Terra/LUNA vs. +225.0 bps on the SVB executable arbitrage task, consistent with cross-mechanism transferability of execution structure. Among broader Tier-B events, peak optical gaps range from ≈ 100 bps (Celsius/3AC, exchange credit) to > 500 bps (Curve/UST, DeFi pool imbalance) and > 800 bps (HUSD issuer failure), all estimated from OHLCV; execution-grade claims require route-complete L2 and are reserved for Tier-A events.

6.7 Oracle Gap Decomposition

The 260 bps gap on the executable arbitrage task decomposes into three components (component 1 derived directly from test-split

results; components 2–3 estimated from the executable-rate sensitivity analysis). First, false-positive cost: for PriceBasis10bps, 581 FP trades at -38.7 bps each produce $-22,484$ bps of total losses, the dominant quantifiable drag for any threshold-based strategy. Second, timing error (estimated 40–80 bps): entering 1–2 minutes late shifts execution to a lower-depth regime, and the depth-withdrawal signature in the 30-minute pre-window is the signal a future model must capture earlier. Third, sizing error (estimated 40–100 bps): the executable rate falls from 2.88% at \$10K to 1.62% at \$50K, pricing out 44% of windows above the retail ceiling. Closing the gap requires cross-mechanism stress training combined with earlier timing entry, not larger notional.

6.8 RL Diagnostic: Density Is Necessary but Not Sufficient

Explicit positive-label supervision beats RL reward optimisation by 111.7 bps at identical training density (17%): this is the paper’s sharpest architectural finding. The three-way ladder below establishes density as necessary but explicit supervision as additionally required. We evaluate the PPO-GRU agent (§5.2) trained on Terra/LUNA stress windows and applied cross-mechanism to the USDC/SVB test split without retraining.

Table 10: RL policy vs. supervised baselines (basis_usdc_1m_gt10bps, SVB test split).

Model (training)	Net bps	Trades
Oracle (hindsight ceiling)	+162.2	315
Meta-labeling (Terra/LUNA*)	+82.5	397
PPO-GRU unconditioned (Terra/LUNA*)	<i>degenerate</i>	0
PPO-GRU conditioned [†] (Terra/LUNA*)	−29.2	919
GRU supervised (calm-ctrl)	−239.0	656
MLP-Window supervised (calm-ctrl)	−340.7	1,052

Terra/LUNA train, SVB eval. [†]Conditioned on primary-signal windows ($|b_{USDC}| > 10$ bps, 17% pos. rate).

The unconditioned PPO-GRU converges to degenerate NoTrade: it makes zero entries despite AUROC 0.74. The 2.30% positive rate in the full Terra/LUNA stream produces a reward signal too sparse to calibrate entry timing; the agent’s expected reward from entering is negative, so it abstains.

A conditioned PPO-GRU, trained only on primary-signal windows ($|b_{USDC}| > 10$ bps, mirroring the meta-label training distribution at 17% positive rate), *does* trade: 919 entries, AUROC 0.87, net -29.2 bps (Table 10). This rules out training sparsity as the sole failure mode. The conditioned RL agent avoids degenerate NoTrade but cannot match meta-labeling (+82.5 bps, 111.7 bps gap), indicating that *explicit positive-label supervision* (which meta-labeling provides via the secondary classifier) outperforms RL reward optimisation at the same training density. The mechanism is qualitative: the meta-labeling classifier receives direct binary supervision ($y=1$ for profitable windows), whereas the RL agent must infer the profitable subset through a reward signal negative on 83% of entries—the supervision is categorically cleaner, not just denser. The 919-trade overtrading pattern (vs. the oracle’s 315) is diagnostic: policies failing from insufficient training converge to NoTrade, not systematic false-positive entry, suggesting a timing-transfer mismatch rather than a fundamental RL limitation. Multi-mechanism training is the productive next step.

7 Limitations

Data scope. The BTC-USDC buy leg uses kline-proxy depth for the SVB window; the 20% haircut sensitivity confirms robustness. Extending Tier-A coverage to AMM or exchange-credit events requires route-complete L2 archives for those venues.

Model scope. The model ladder covers tabular classifiers, sequence models, and cross-mechanism PPO-GRU variants. The conditioned RL result (−29.2 bps, 919 trades) may reflect overfitting to Terra/LUNA’s specific depth-withdrawal timing rather than a fundamental failure of sequential optimisation; resolving this requires additional cross-mechanism pairs or longer training runs. Positive example density and explicit positive-label supervision are jointly necessary for positive net returns (§6.8).

Market-impact feedback. Unlike reactive-simulator approaches such as Olby et al. [23], in which agents observe how their own order flow consumes book depth, the PPO-GRU trains on historical replay and cannot model this feedback loop. In a thin-book stress regime this systematically miscalibrates entry sizing: the agent observes book depth as it was, not as it would be after its own orders clear liquidity. This is a candidate explanation for the 919-trade overtrading pattern observed at identical training density to meta-labeling. Reactive-simulator training is the productive next step for closing the timing component of the oracle gap, combining the cross-mechanism initialisation protocol shown here with an environment that updates depth in response to agent actions.

8 Conclusion

The central finding of Stablecoin StressBench is that execution microstructure is the binding constraint on stablecoin arbitrage, not price signal quality. A 34.3% optical dislocation rate collapses to 2.88% once order-book depth and transaction costs are applied. No model trained on calm-market data closes this gap. Cross-mechanism training on a different stress event is the empirically validated path to positive net returns (+82.5 bps, 50.8% oracle capture). No calm-control-trained model earns positive returns on the executable arbitrage task; LightGBM (+17.2 bps, basis task only) is the sole non-oracle positive result, 145 bps below the oracle ceiling. The oracle gap decomposition (§6.7) identifies false-positive cost as the dominant quantifiable drag and timing as the primary structural barrier.

The RL experiments (§6.8) sharpen the benchmark implication: **positive example density is necessary but not sufficient; explicit supervision of the positive subset is the additional requirement.** The unconditioned PPO-GRU degenerates (2.3% density too sparse). The conditioned PPO-GRU trained on the same 17% primary-signal windows as the meta-label does trade (−29.2 bps, 919 trades) but cannot match meta-labeling (+82.5 bps), leaving an 111.7 bps gap attributable to meta-labeling’s explicit positive-label supervision rather than RL reward optimisation.

Benchmark deliverables. The anonymised repository (URL redacted for review) ships five concrete artefacts: (1) per-minute executable labels for all six benchmark windows; (2) the VWAP book-walking label-construction code with route-level depth provenance; (3) the four nested feature sets (price_only through price_book_settle); (4) the hindsight oracle implementation and

full model-ladder evaluation harness; and (5) the cross-mechanism meta-labeling and PPO-GRU training scripts. Calm-control and SVB windows are reproducible from the public Binance Vision archive; route-complete L2 requires a Tardis subscription.

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